

Ryan Wetzstein

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EDUCATION

North Dakota State University - Fargo, North Dakota Expected Fall 2027

Bachelor of Science in Computer Science | Minor: Artificial Intelligence GPA: 3.444

- Relevant Coursework: Introduction to Artificial Intelligence, Software Projects Capstone 2026, Operating Systems Concepts, Computer Organization and Architecture, Software Development with Frameworks
- Dean's List: Fall 2023, Fall 2024, Spring 2025, Spring 2026

TECHNICAL SKILLS

- Languages: Java, Python, SQL, TypeScript, C#
- Frameworks/Tools: React, Vite, PostgreSQL, Tauri, TensorFlow/Keras
- Concepts: Agentic Development, OOP, Agile/Scrum, Relational Databases, Neural Networks, Machine Learning

PROJECTS

Nexis - Open-source AI-native terminal & IDE May 2026 – Current

Rust, Tauri, React, TypeScript, xterm.js, CodeMirror 6 | nexusdev.org

- Directed an agentic-assisted rewrite of a forked open-source terminal into a full cross-platform IDE, authoring 285+ commits over 4 months across a Rust/Tauri backend and React/TypeScript frontend.
- Authored the React 19 / TypeScript frontend, xterm.js terminal, CodeMirror 6 editor, and file explorer, and designed a sandboxed, approval-gated AI agent system spanning 12+ LLM providers, with OS-keychain credential storage and SSRF/DNS-rebinding protections.
- Specified and integrated the Rust/Tauri backend, native PTY, LSP (go-to-definition, diagnostics, refactors), and a DAP step-through debugger. Packaged application remains under 10MB with zero telemetry.

2026 NDSU Capstone - FarmQA January 2026 – May 2026

React, Vite, Microsoft Azure Blob Storage, Functions, SQL Server | farmqa.com

- Built a local backend management application for FarmQA's internal use to view and monitor Azure Blob Storage data and metrics.
- Worked in a team of 4 as part of the NDSU Capstone program; delivered a usable, client-approved product and presented it at the 2026 Senior Design Expo.

Music Genre CNN - Audio classification with CNNs Spring 2026

Python, TensorFlow/Keras, librosa, NumPy | github.com/rwetz/music-genre-cnn

- Built an end-to-end audio classification pipeline converting raw waveforms into normalized 128x128 mel-spectrogram images with librosa, split 80/10/10 into train/validation/test sets.
- Trained and compared three CNN architectures: a 3-block baseline, a 4-block variant with batch normalization, and a lightweight global-average-pooling model, evaluating each with training curves and confusion matrices and automatically checkpointing the best performer.

WORK EXPERIENCE

Lemhi Technologies - Startup Company May 2026 – August 2026

Full-Stack Development Intern | Remote

- Collaborated directly with startup founders on product scope, use-case definition, and system selection for implementation, including creating user-flow diagrams for proposed solutions.
- Built [arc-gis-harvester](#), a Python web scraper that collects public agriculture and land GIS data to construct a dataset underpinning the company's product.
- Participated in regular strategy meetings with founders, contributing directly to decisions on product scope and UI/UX direction.